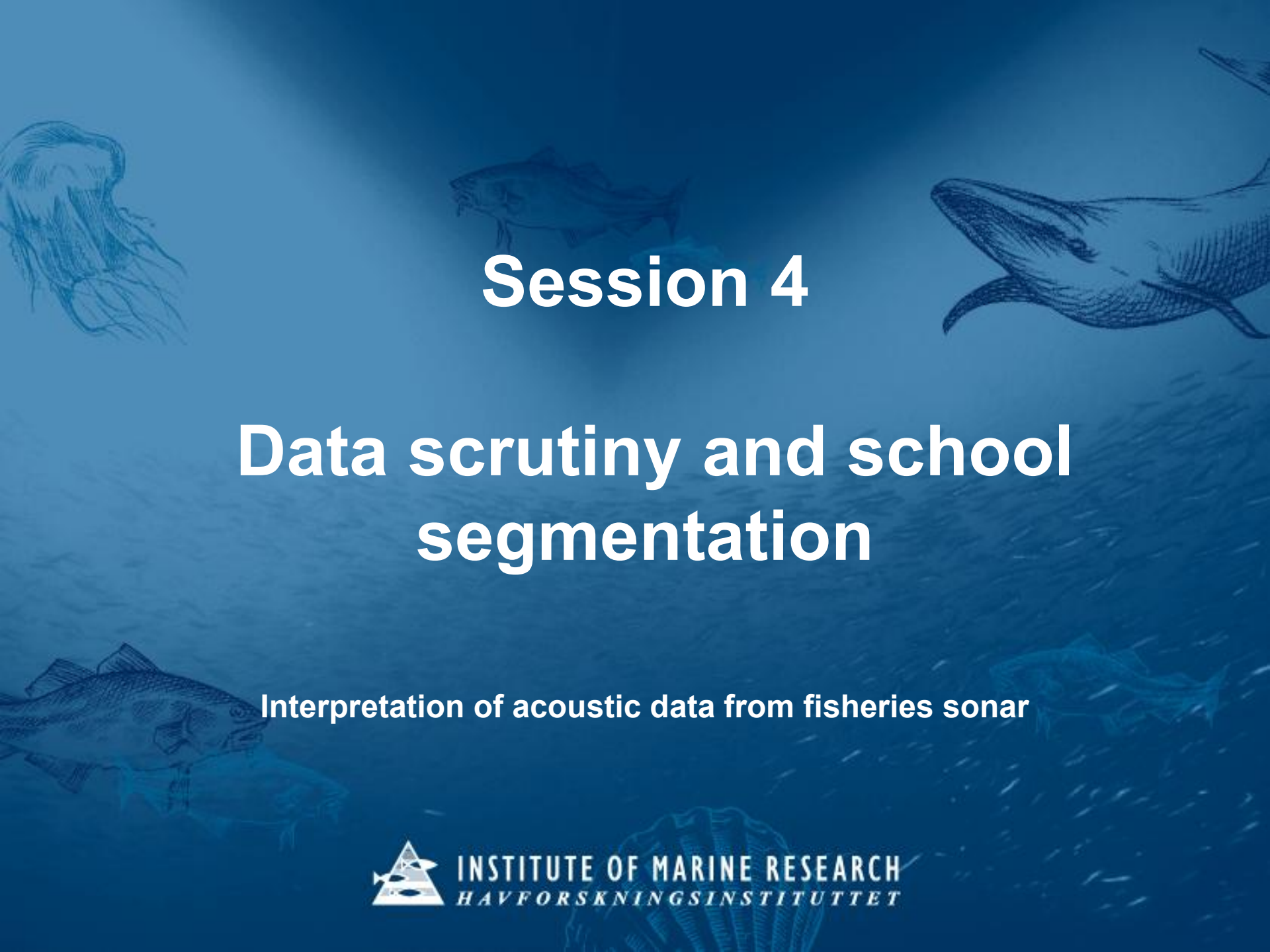




Session 4

Data scrutiny and school segmentation

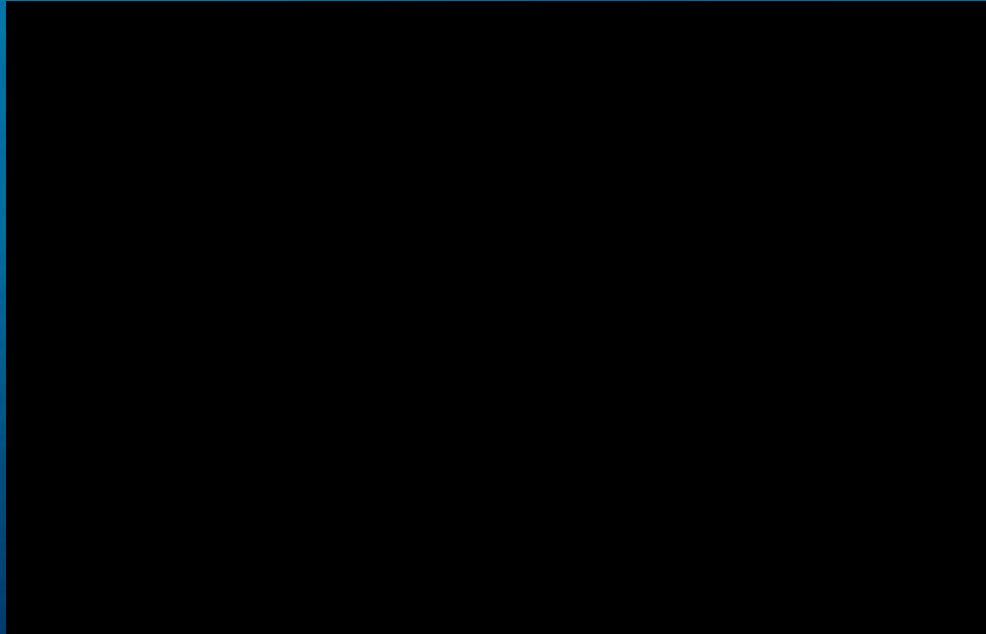
Interpretation of acoustic data from fisheries sonar

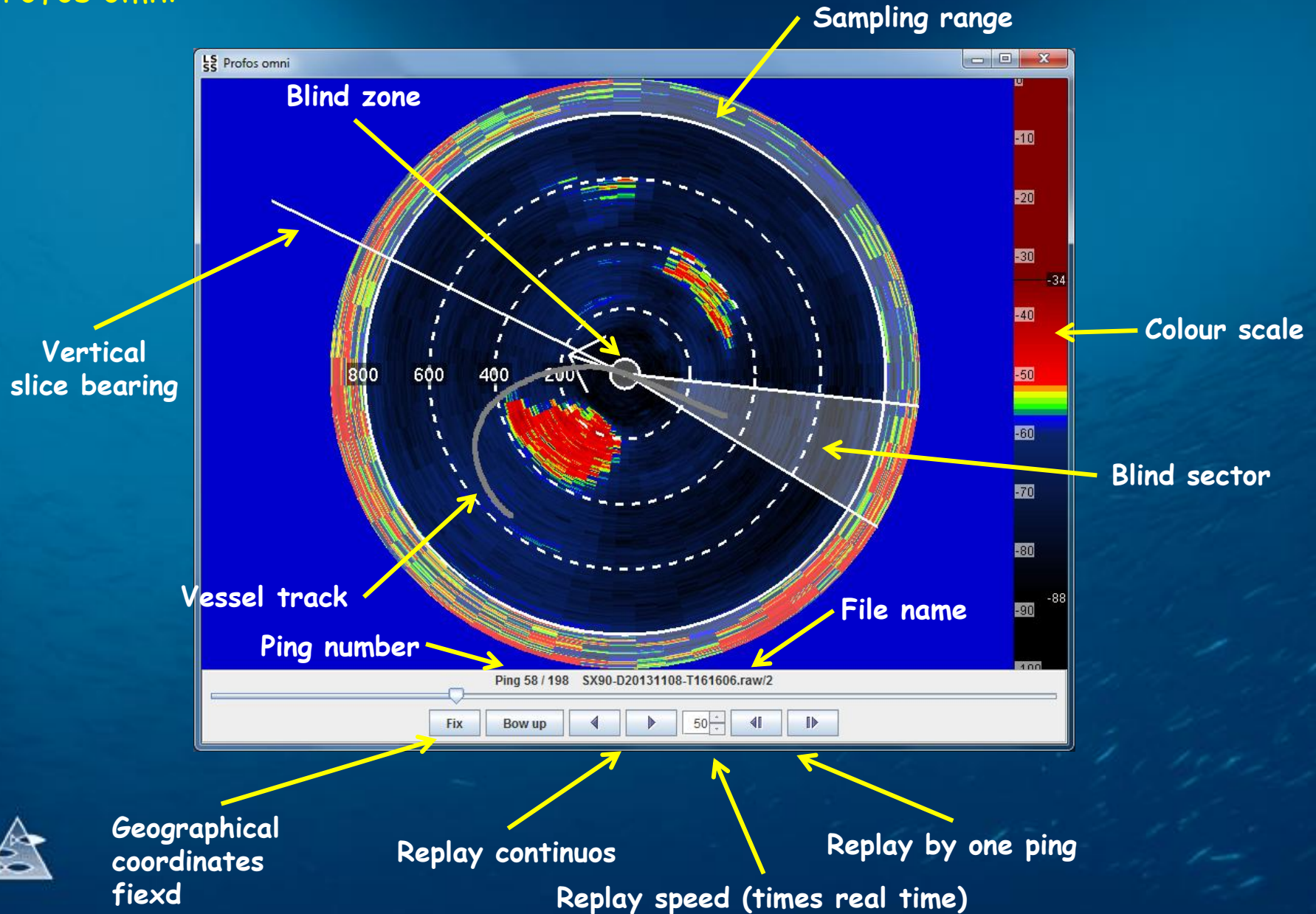


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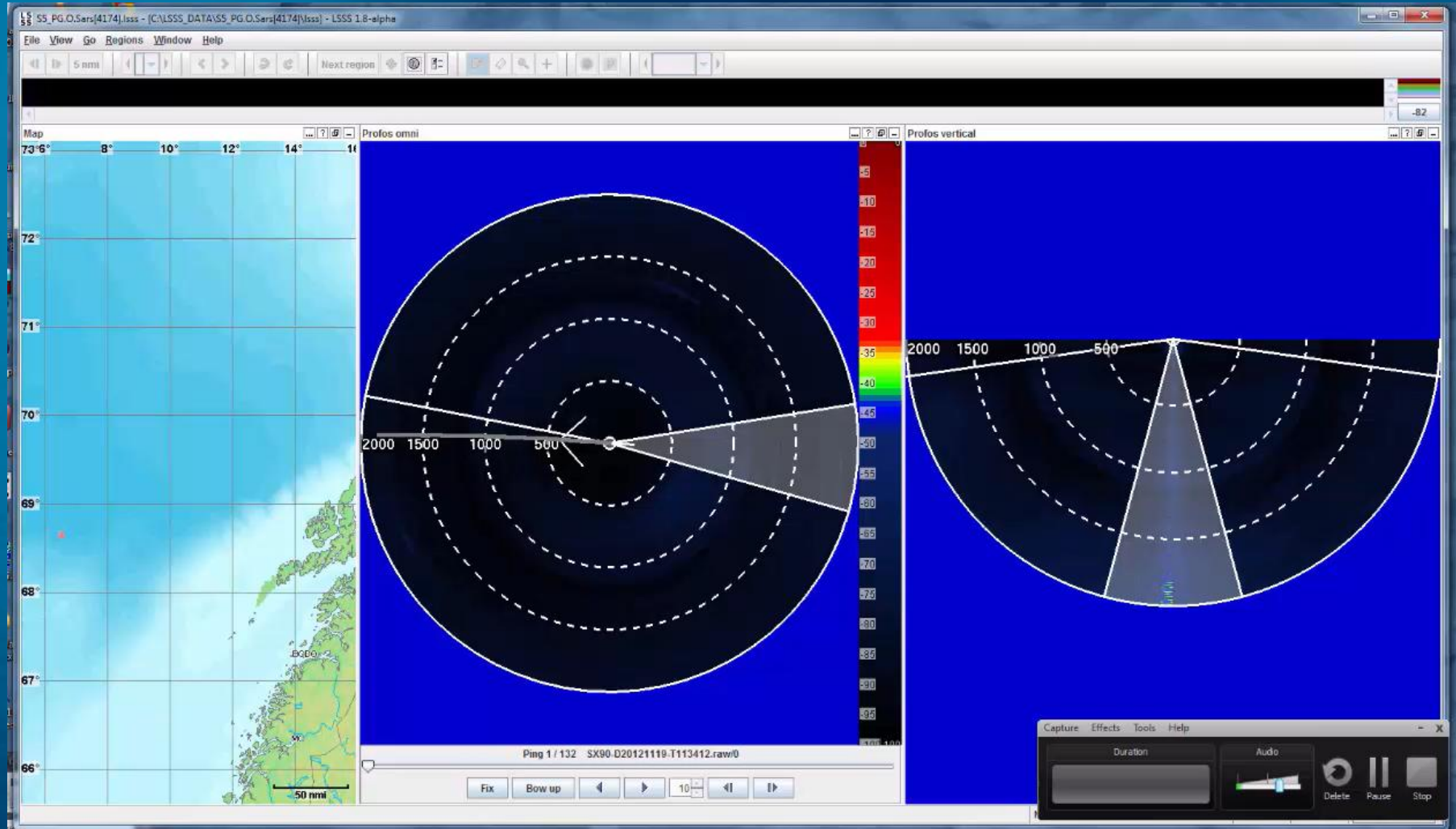
Start working with Processing system for omni directional fisheries sonar (Profos)

2. Temporary and preliminary inspection of sonar data
 - No need to create a "Survey"
 - Other data sources can be loaded
 - No database is created, and no data stored to database





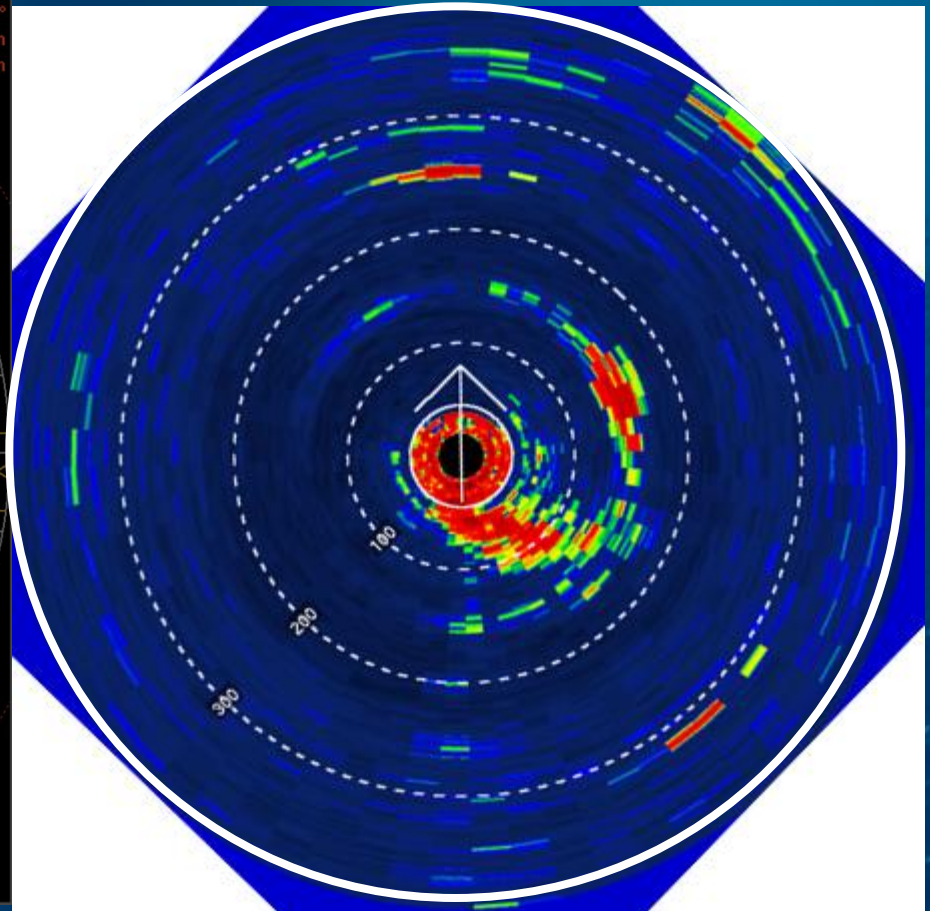
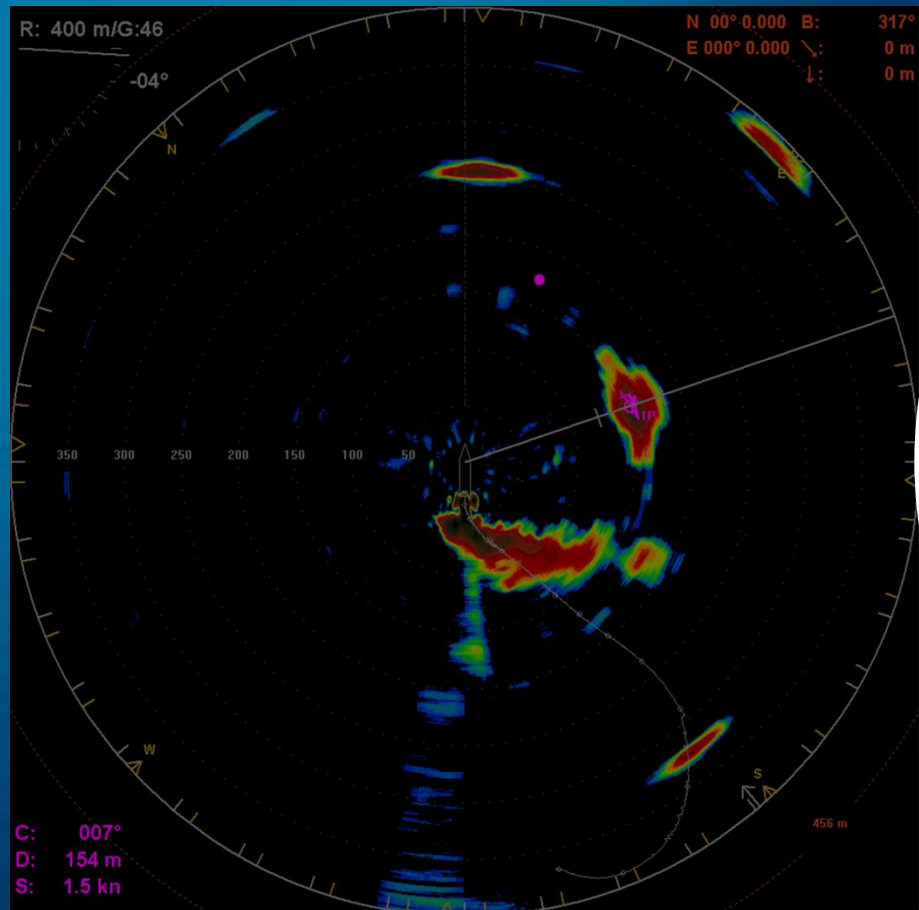
Sonar module windows



Sesion_4_3

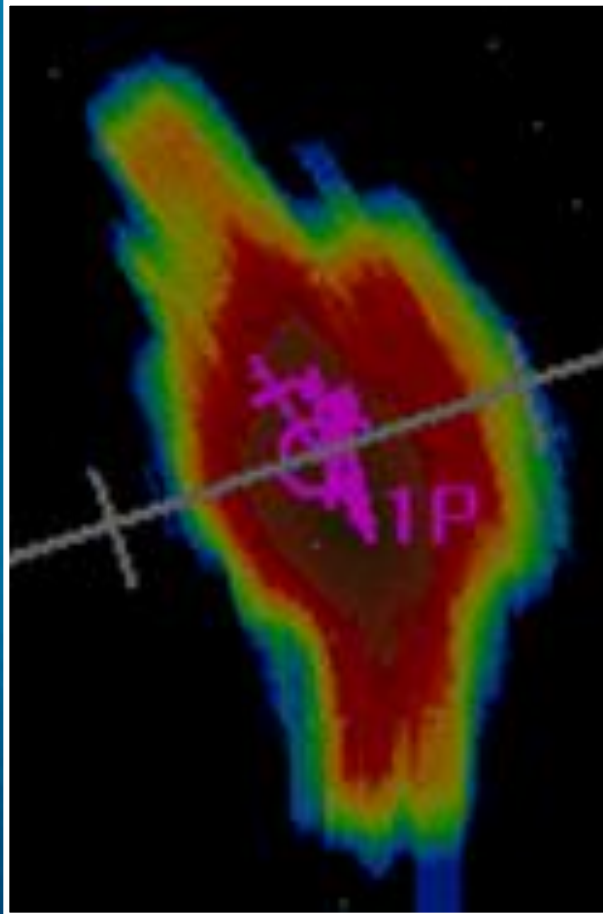


Sonar display and Sv computed from raw data

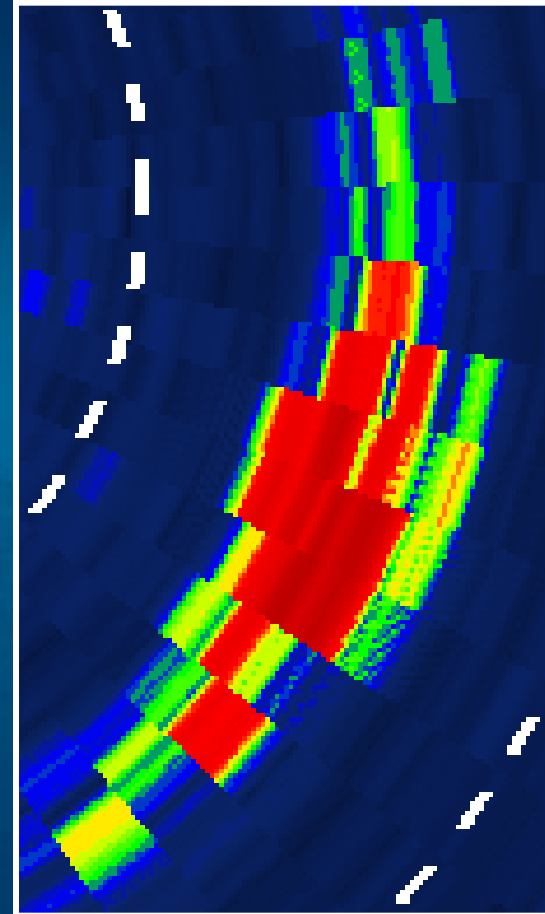


Detail of a school in sonar display and school in processed raw data

Sonar screen dump



Raw data in PROFOS



Sonar display shows highly filtered data with the purpose to enhance the school and reduce the scattering from unwanted targets like surface reverberation or bottom

Schools scattering is enhanced so they can be detected at longer ranges and with higher resolution

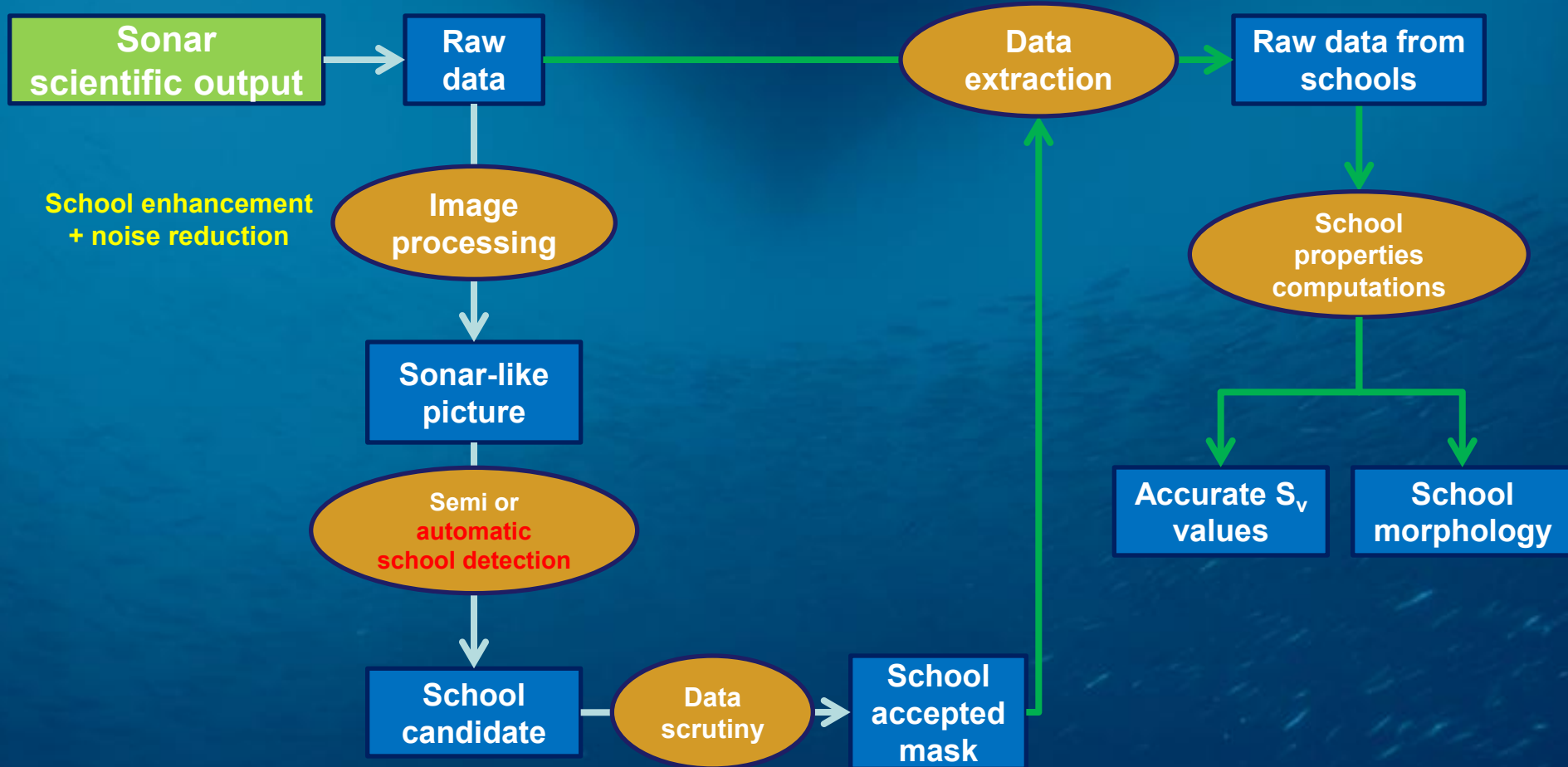
Raw data from scientific output has not been subject to any processing

challenge

Get a "nice picture" (low noise and enhanced schools)
and correct Sv values

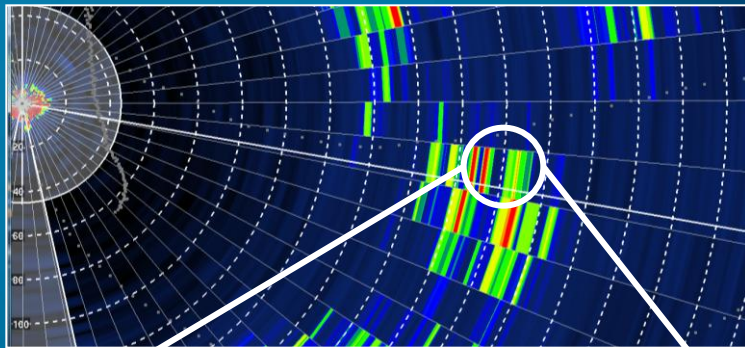


Sonar data processing in PROFOS



School enhancement + noise reduction

Median filter along each beam

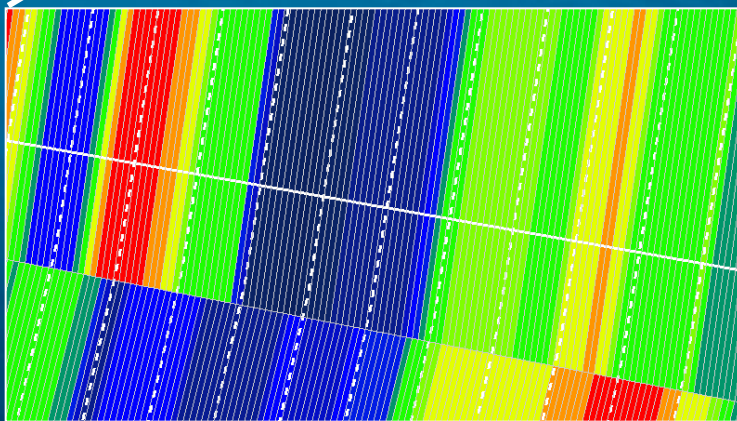


Beam 1

Beam 2

Beam 3

Beam 4



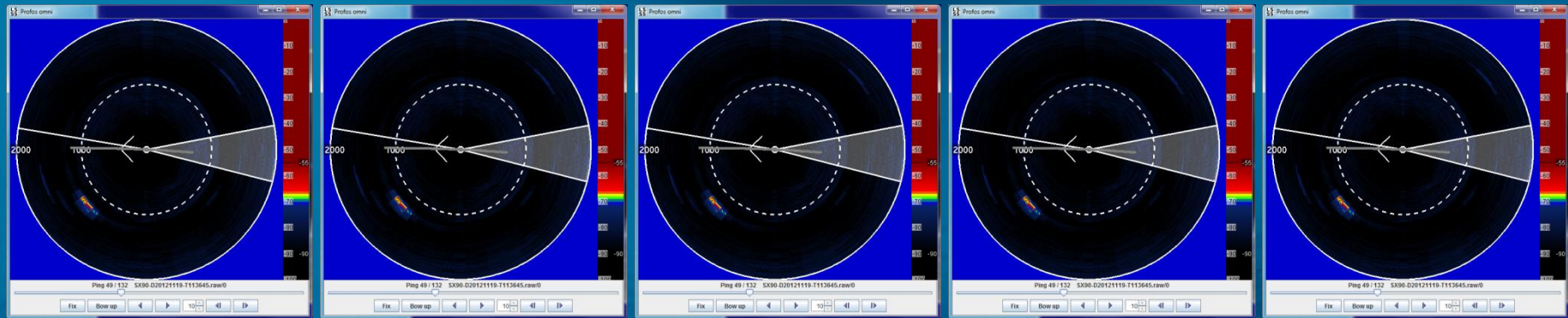
Default:
Median filter for 21 samples in each beam

Samples along a beam
(ca. 2000 samples)



School enhancement + noise reduction

Median filter between consecutive pings



Ping 1

Ping 2

Ping 3

Ping 4

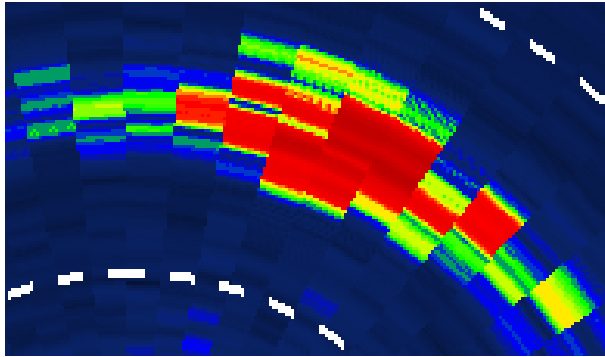
Ping 5

Default: median filter between 5 pings with geographical correction



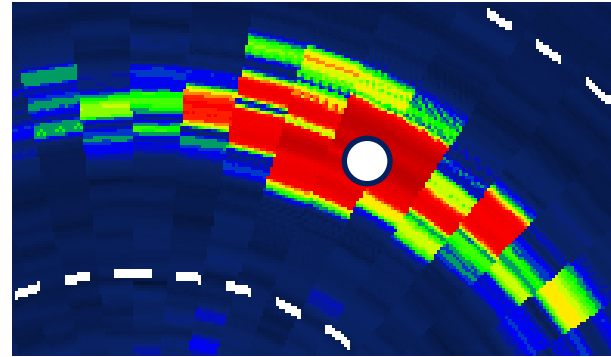
Semi-automatic school segmentation

1. School selection

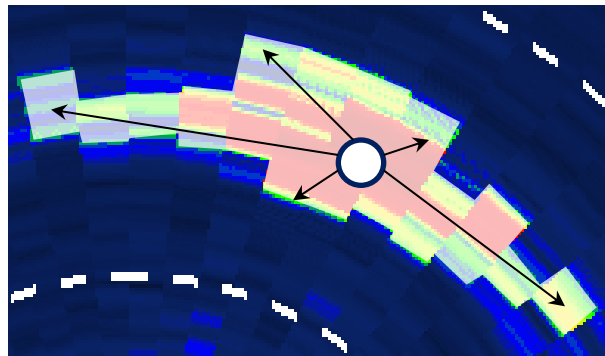


Automatic school detection

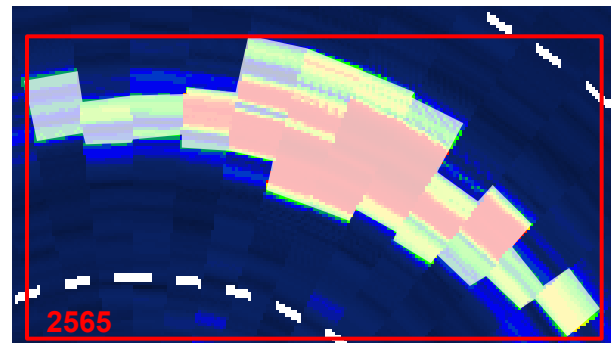
~~2. Manual seeding~~



3. Automatic school segmentation

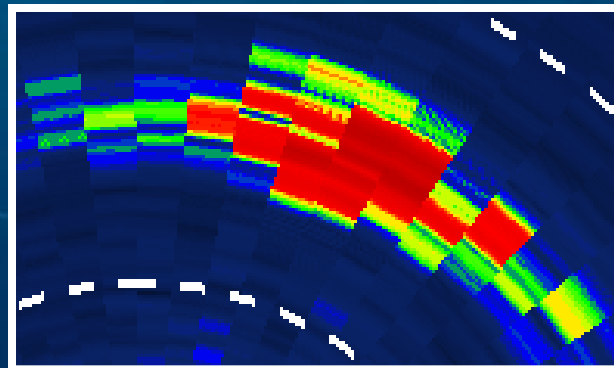


4. School ID with properties



Characteristic of the schools in sonar data

- Backscattering above the background level
- Have heterogeneity in strength with higher values in centre
- Should be observed in several consecutive pings
- Should have a spatial displacement in consecutive pings
- Have a "realistic" dimensions and oval shape



Segmentation parameters

LS

SS

Grow parameters

×

Omni grow parameters

UseAdaptiveThreshold

☒

AdaptiveThreshold

8

[dB]

Min value for growing = range dependent trend + AdaptiveThreshold

MinValue

-45

[dB]

Min value for growing

MaxValue

-20

[dB]

Max value for growing

MaxPingFromSeed

10

[#]

Max pings from seed

MaxMissingPings

1

[#]

Max number of consecutive pings where school can be missing

MinMaxDeltaSv

2

[dB]

Regions are rejected if for all samples $S_v < \text{min value for growing} + \text{MinMaxDeltaSv}$

MinArea

100

[m²]

Regions are rejected if area < MinArea

MaxArea

30000

[m²]

Regions are rejected if area > MaxArea

MaxAspectRatio

10

[–]

Regions are rejected if $\text{sizeAlongRing} / \text{sizeAlongBeam} > \text{MaxAspectRatio}$

MinPingCount

7

[#]

Schools are rejected if they span less than MinPingCount number of pings

DiskRadius

100

[m]

Radius used when creating school disk

OK

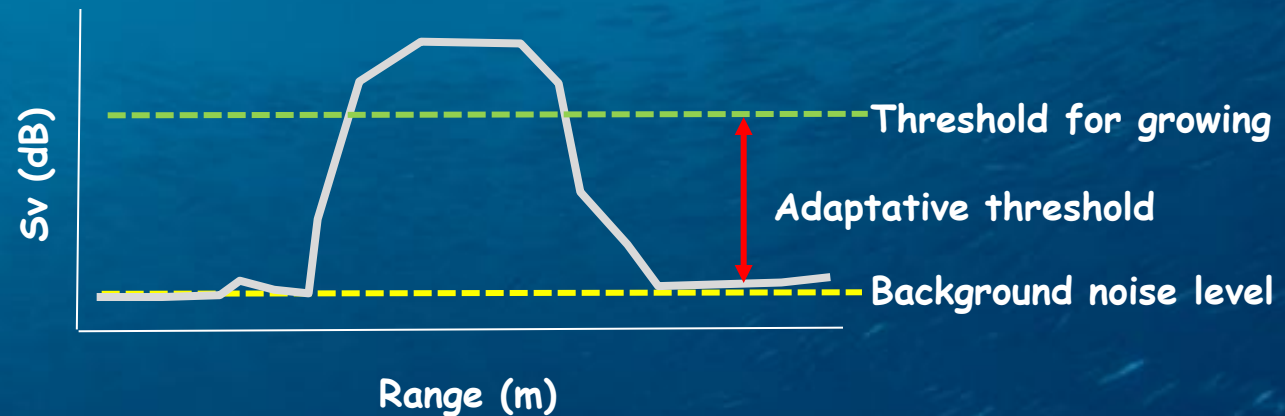
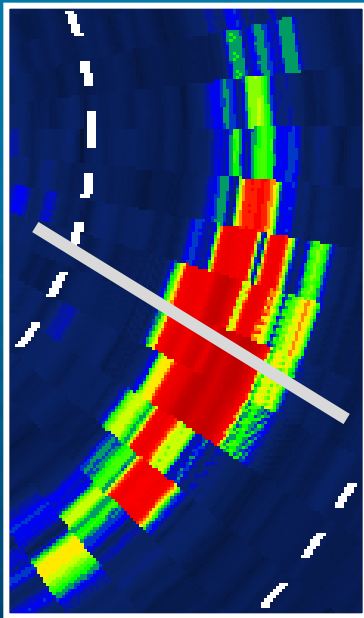
Cancel



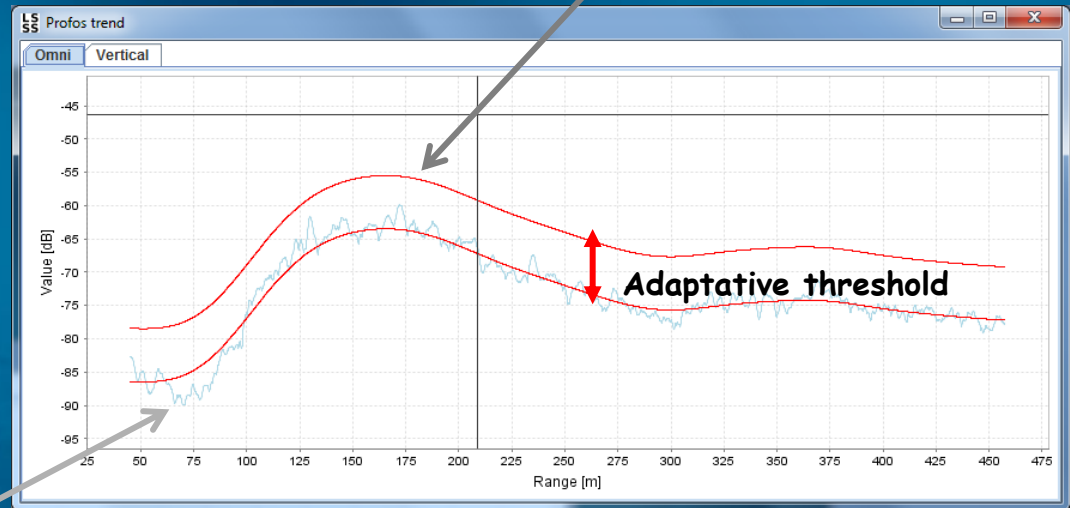
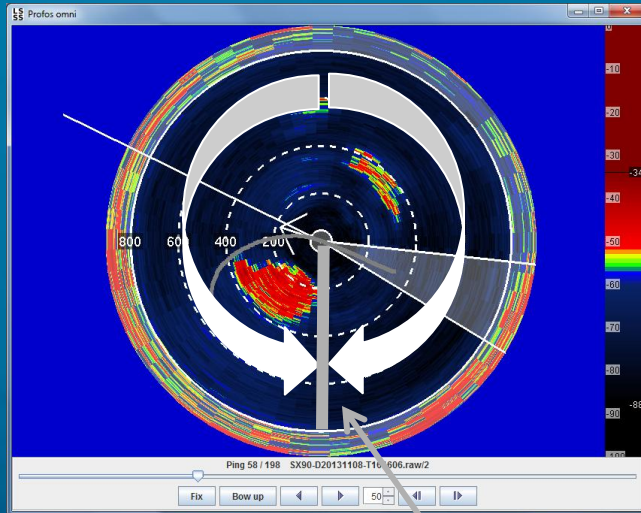
Adaptative threshold

Not fixed, changes every ping
based in background noise level

Level above the background noise
that could be clasified as a school



Minimum value for growing = range dependent trend + **Adaptative threshold**



Threshold for growing
(minimum value for growing)

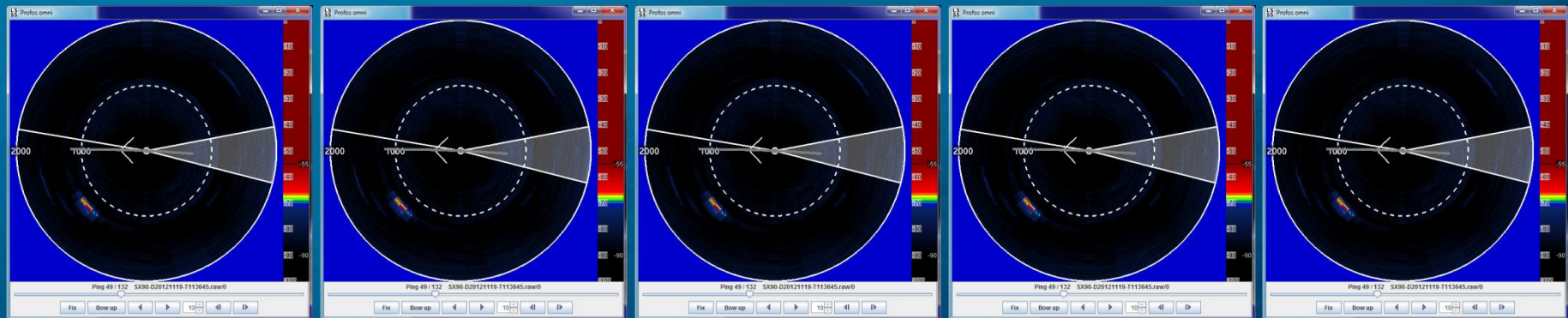
Average of all samples by ring
(range dependent trend)



MaxPingFromSeed

Maximum number of pings back and forward from seed ping to segment school

$\text{MaxPingFromSeed} = 2$



Ping 1

Ping 2

Ping 3

Ping 4

Ping 5

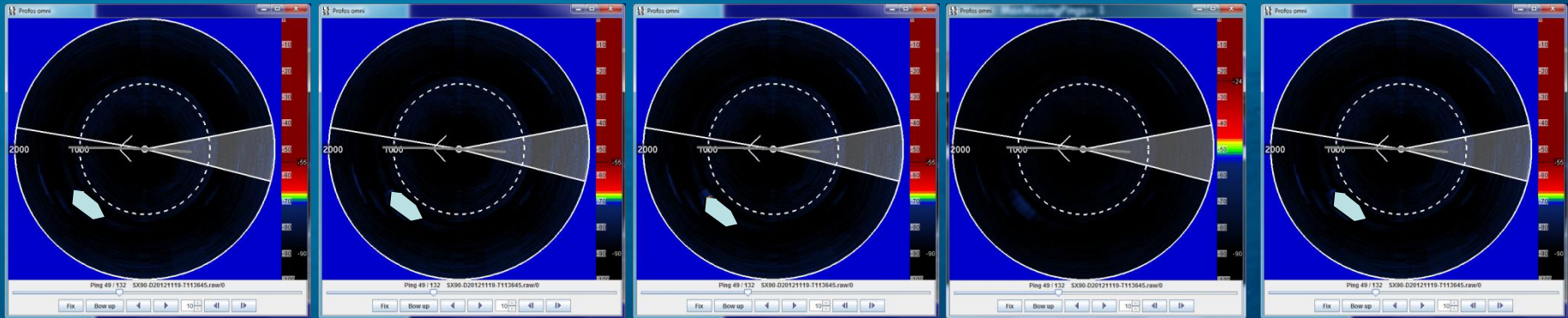
Seed ping



MaxMissingPings

Maximum number of pings that can be missing when growing a school

MaxMissingPings= 1



Ping 1

Ping 2

Ping 3

Ping 4

Ping 5

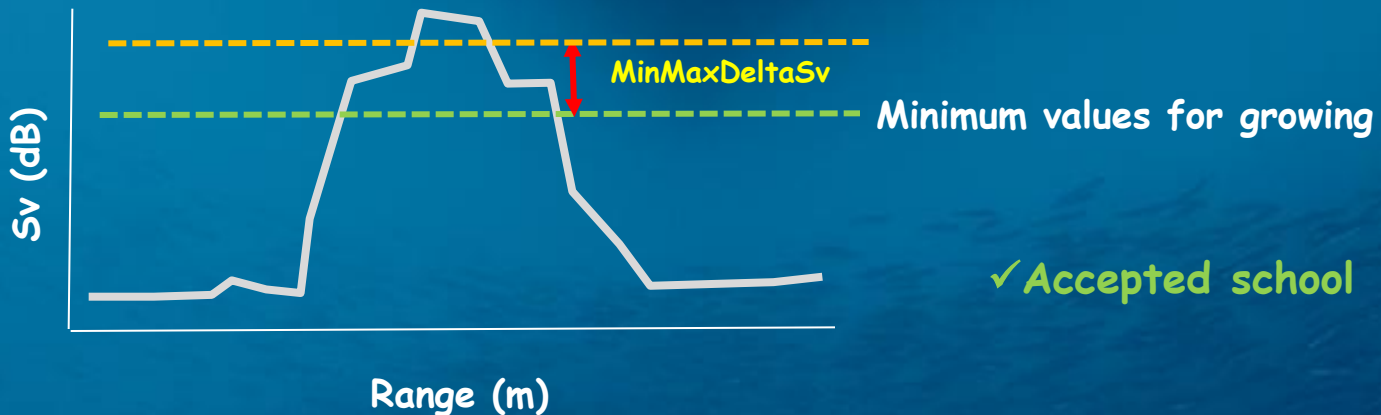
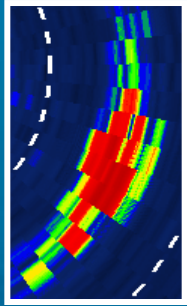


Ping with no school detected



MinMaxDeltaSv

Parameter which indicates that some of the samples of the school candidates have to be stronger than the minimum values for growing



MinArea

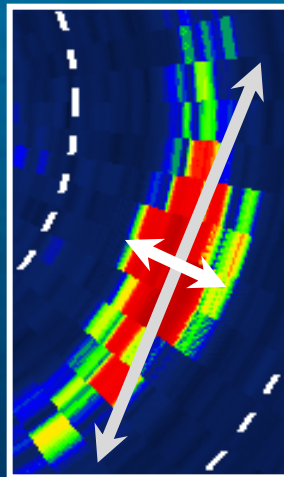
Minimum area of school (m^2), if smaller, school candidate is rejected

MaxArea

Maximum area of school (m^2), if larger, school candidate is rejected

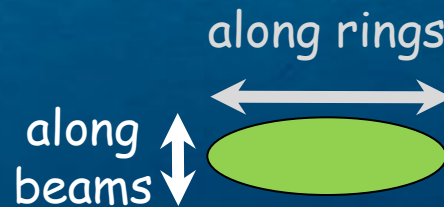
MaxAspectRatio

Maximum aspect ratio between size along rings / along beams . If larger school candidate is rejected



along rings

along beams



✓ Accepted school

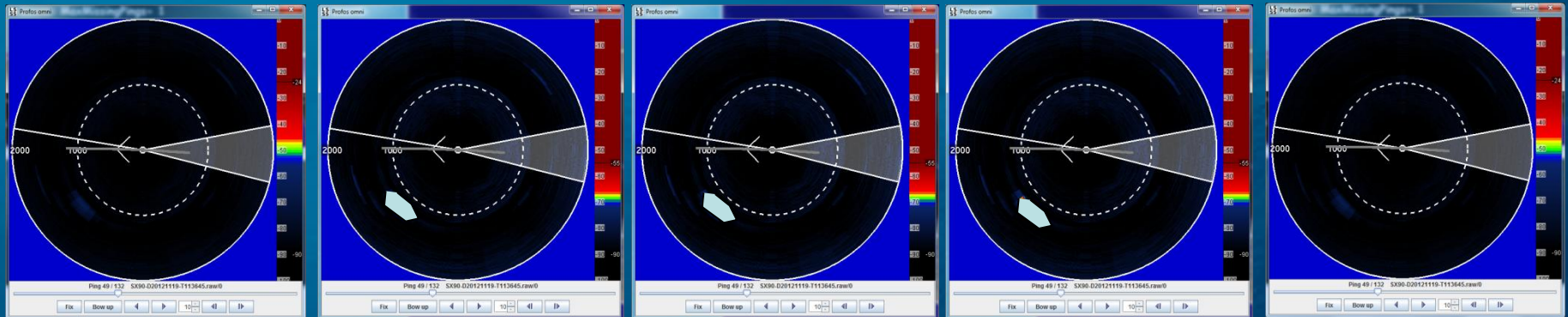


✗ Rejected school



MinPingCount

Schools are rejected if they are not detected a minimum number of pings



Ping 1

Ping 2

Ping 3

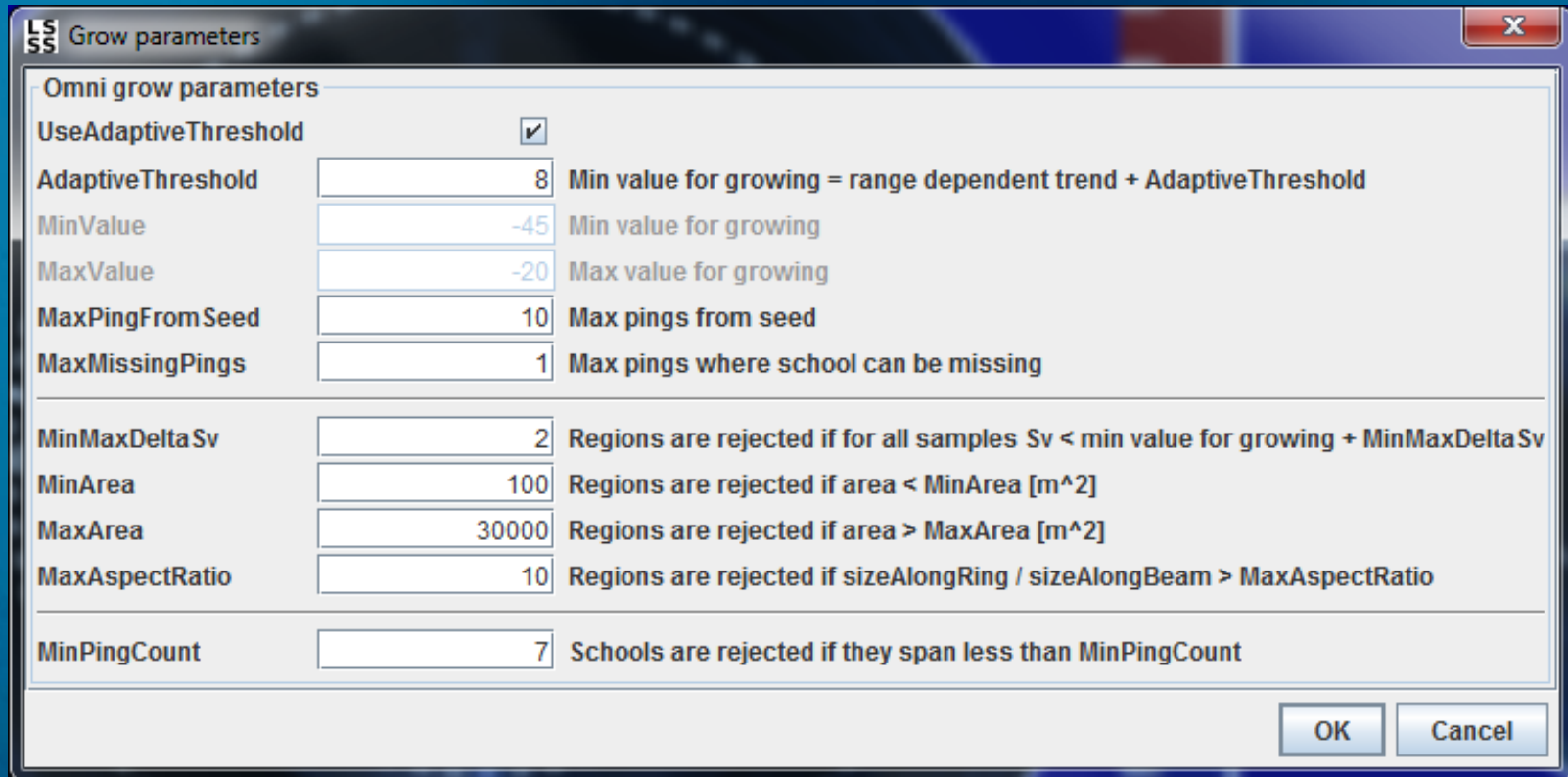
Ping 4

Ping 5

MinPingCount=3



Default parameters



The screenshot shows a software window titled "Grow parameters" with a close button (X) in the top right corner. The window contains a list of parameters for growing schools, each with a text input field and a description. The parameters are:

Omni grow parameters		
UseAdaptiveThreshold	☒	
AdaptiveThreshold	8	Min value for growing = range dependent trend + AdaptiveThreshold
MinValue	-45	Min value for growing
MaxValue	-20	Max value for growing
MaxPingFromSeed	10	Max pings from seed
MaxMissingPings	1	Max pings where school can be missing
MinMaxDeltaSv	2	Regions are rejected if for all samples $S_v < \text{min value for growing} + \text{MinMaxDeltaSv}$
MinArea	100	Regions are rejected if area $< \text{MinArea [m}^2\text{]}$
MaxArea	30000	Regions are rejected if area $> \text{MaxArea [m}^2\text{]}$
MaxAspectRatio	10	Regions are rejected if $\text{sizeAlongRing} / \text{sizeAlongBeam} > \text{MaxAspectRatio}$
MinPingCount	7	Schools are rejected if they span less than MinPingCount

At the bottom right of the window are "OK" and "Cancel" buttons.

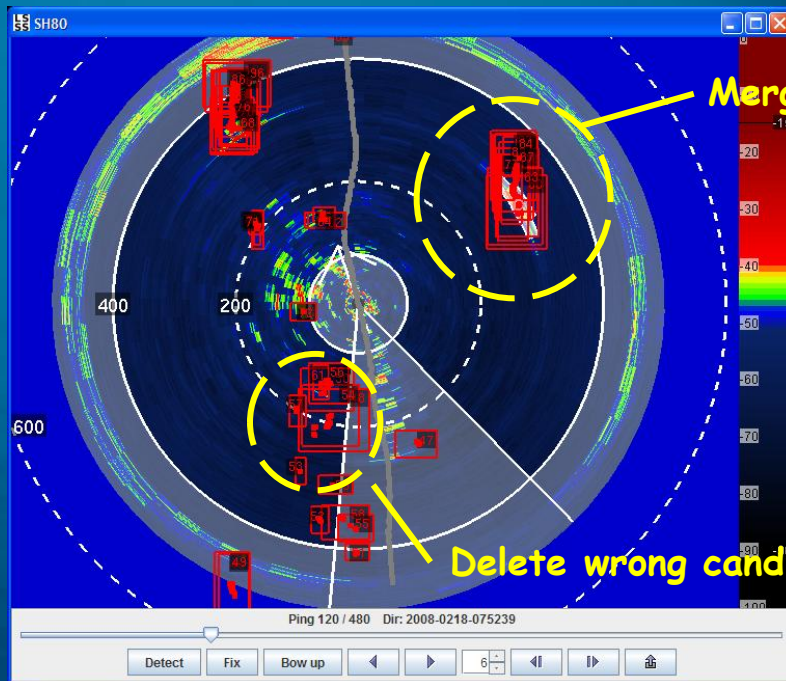
**Parameters need to be adjusted for each survey condition
(sonar transmission and fish behaviour)**



Need to quality control the automatic school detection results

Capelin schools winter 2008 (SH80 Libas)

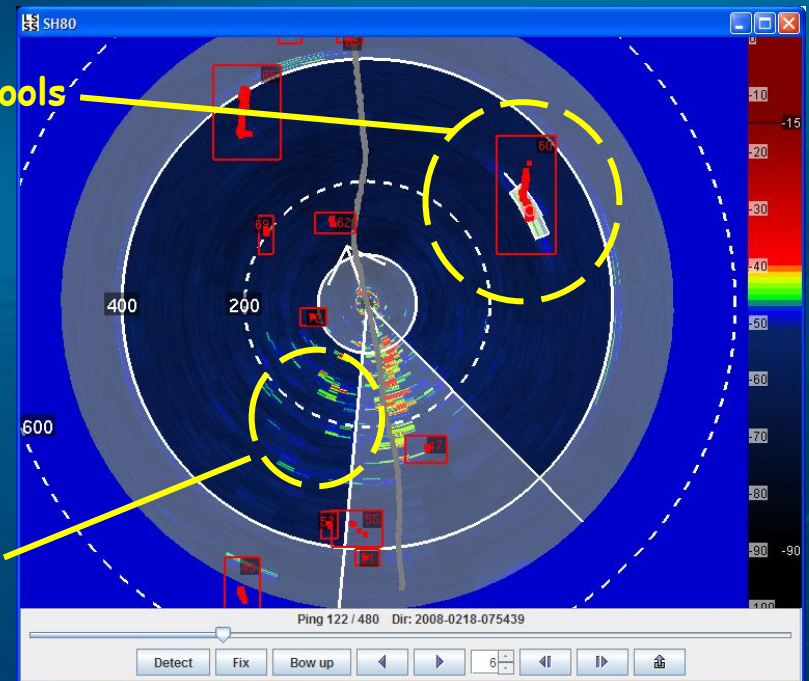
Automatic school growing



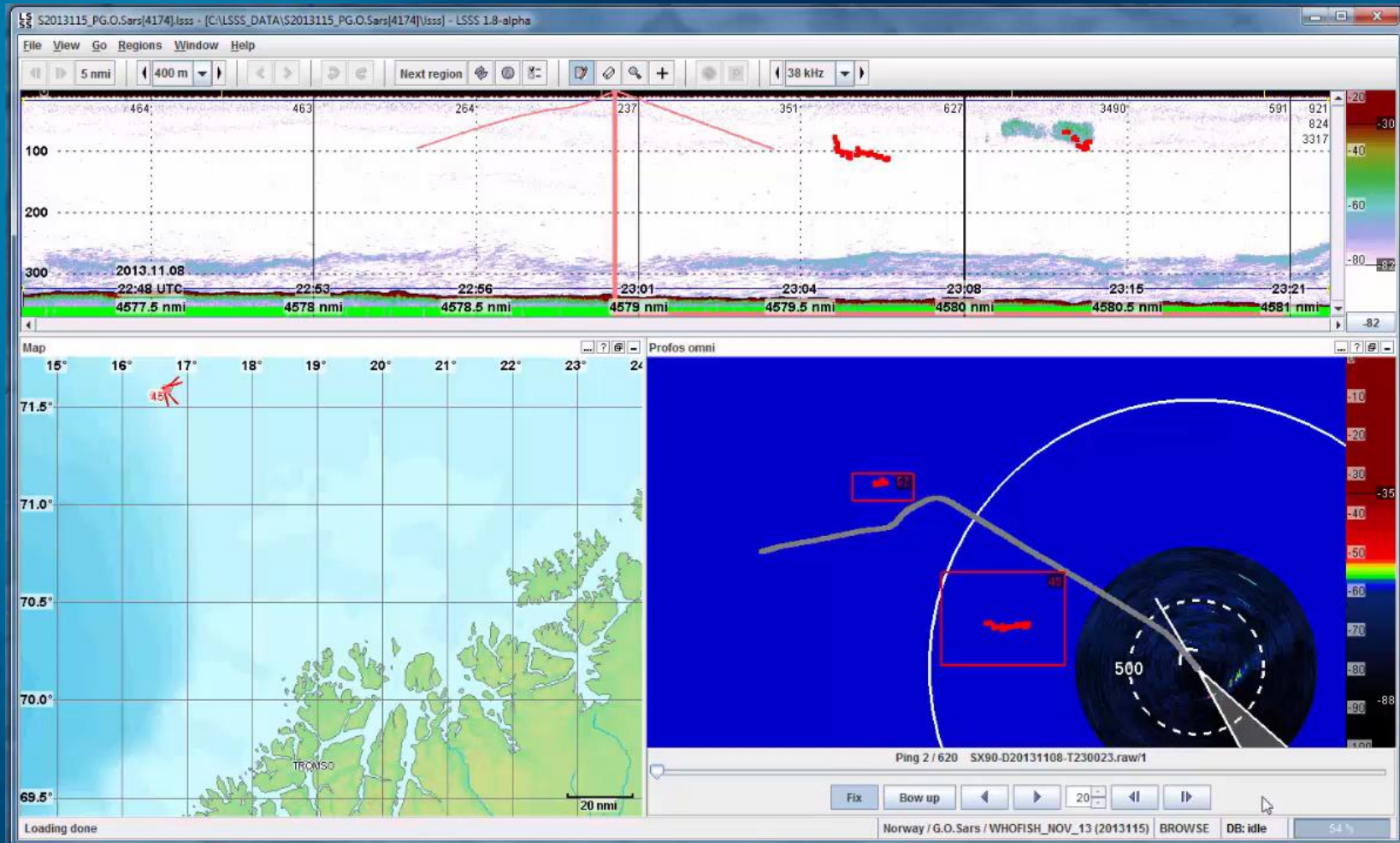
Merge schools

Delete wrong candidates

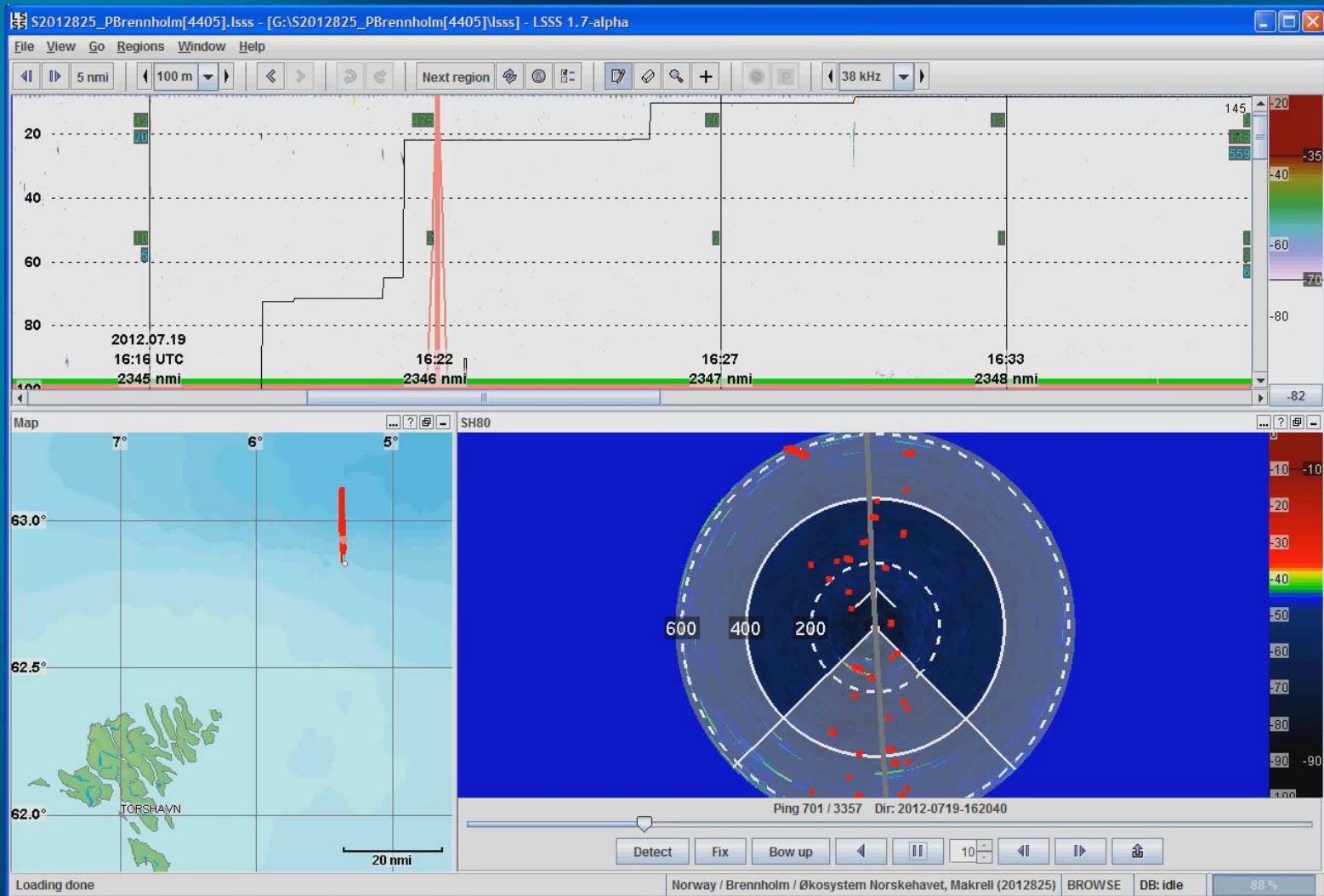
After manual scrutinizing



Example of scrutinized large herring schools winter 2013



Example of scrutinized small herring schools summer 2010



LSSS scrutinized schools with echo

